TDM Radar Target Angle and Velocity Estimation with Forward and Reverse Transmission Patterns

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Abstract

Time division multiplexing multiple-input multiple-output (TDM-MIMO) radar is widely used in vehicle applications, but it suffers from the problem of velocity and angle coupling when measuring moving targets. In order to eliminate the influence of TDM on parameter estimation, it is necessary to design a new transmission pattern and corresponding processing method. In this paper, a forward and reverse transmission pattern is proposed together with the parameter estimation method. The forward and reverse transmission are used to obtain the sum and difference of target velocity and angle phases respectively, and the independent parameters are solved from the coupled phases. Different from the commonly adopted random TDM schemes, the transmission pattern of proposed forward and reverse transmission is determined, and the estimation algorithm mainly involves only two-dimensional fast Fourier transform (FFT) on uniformly sampled data matrixes. The asymptotic estimation accuracy and the multi-target resolution is equivalent to that of multiple-input single-output (MISO) radar with the same aperture, which means that the proposed method overcomes the coupling problem in TDM radar.

1 Introduction

Time division multiplexing multiple-input multiple-output (TDM-MIMO) radar is widely used in automobiles because of its simple structure and low cost [1], [2]. Although the radar aperture is greatly expanded with multiple transmit antennas, the Doppler coupling problem caused by different transmitting times hinders the detection of high-speed targets, which is a key issue in automotive applications [3].

Early researches eliminated the coupling on angle measurement by estimating the velocity first and measuring the angle after compensation [4], [5]. In traditional TDM radar, the target velocity estimated by multiple transmits of one virtual antenna is used for compensation directly [6]. However, TDM reduces the velocity unambiguity range significantly, and the wrong velocity estimation will invalid the compensation. Later researches established the TDM signal model of moving target and proposed various joint estimation methods [7], [8]. Although joint estimations decoupled the velocity and angle, the speed measurement is still restricted by the reduced unambiguous range, which means that new transmission patterns must be designed [9], [10]. A simple solution is to redesign antenna locations and set overlapping virtual antennas, which can improve the Doppler sampling rate at certain location but at the cost of reducing antenna aperture [11].

On the other hand, many researches have resorted to the compressed sensing algorithms, which randomizes the transmission order and recover the target velocity from under sampled data [12], [13]. Many sparse recovery methods have been tried to estimate the velocity, of which the match filtering of single target is the simplest [14]. This method is easy to implement by arranging data in each range bin into a two-dimensional matrix, supplement the missing data with 0 and performing two-dimensional fast Fourier transform (FFT) [15]. In order to further process multi-target signal, compressed sensing algorithms and matrix completion algorithms are further adopted to reduce the sidelobe caused by random transmission [16]–[18]. In addition to the high complexity, the performance of these algorithms largely depends on the transmission pattern, which may lead to significant side lobes in special cases [19], [20]. The lack of universal and simple transmission pattern design criteria also hinders the application of random transmission.

Consistent with foregoing discussion, the main contributions of this paper are to propose new TDM transmission patterns and corresponding parameter estimation methods. First, a forward and reverse transmission patterns is designed, in which the forward transmission activates the transmitting antennas in spatial order like traditional TDM radars, while the reverse transmission dose the opposite. The two transmissions can be uncoherent, for example, with unknown time intervals. Secondly, different from the previous methods which directly estimate the independent velocity and angle, the proposed algorithm estimates the sum and difference of velocity and angle phases first in forward and reverse transmission respectively. Specifically, the forward and reverse transmission data are organized into matrixes, and the parameters are estimated by two-dimensional Fourier transform. This method has low complexity and reaches the Cramer-Rao bound in case of single target. Its estimation accuracy and resolution are also equivalent to the MISO radar with the same aperture, which means that the proposed method can effectively overcome the coupling problem in TDM radar.

2. Methodology

2.1 TDM Signal Model with Proposed Forward and Reverse Transmission

This paper focuses on the joint estimation of target angle and speed in TDM radar, so the influence of signal bandwidth is not considered and each pulse is sampled only once at each receive antenna. The N transmit and M receive antennas are linear and uniformly spaced and numbered 0 to $N-1$ and 0 to $M-1$ according to their locations. The receive array spacing is $\lambda/2$, where λ is the carrier wavelength, and the transmit antenna spacing is $M\lambda/2$. The pulse repetition time T_p is also constant. If the transmit antenna transmits in order of number 0 to $N-1$, the signals of n th pulse sampled by M receive antennas with K targets are

$$\mathbf{s}_f[n]=\left[s_f[n,0],s_f[n,1],...,s_f[n,M-1]\right]^T, n=0,1,...,N-1, \quad (1)$$

where

$$\begin{aligned} s_f[n,m] &= \sum_{k=1}^{K}\sigma_{k,f}\exp\left(j\left(nM+m\right)\theta_k+jnv_k\right), m=0,1,...,M-1 \\ &= \sum_{k=1}^{K}\tilde{\sigma}_{k,f}\exp\begin{pmatrix} j\left(n-\left(N-1\right)/2\right)\left(M\theta_k+v_k\right) \\ +j\left(m-\left(M-1\right)/2\right)\theta_k \end{pmatrix}, \end{aligned} \quad (2)$$

where the $\sigma_{k,f}$ is the signal complex amplitude of the k th target in forward transmission, and

$$\tilde{\sigma}_{k,f}=\exp\left(\left(N-1\right)jv_k/2+\left(MN-1\right)j\theta_k/2\right)\sigma_{k,f}, \quad (3)$$

is the complex amplitude with reference to the centre of array and time. The superscript T means transpose, and j is imaginary unit. The phase shift between the receive antennas caused by the angle of the k th target is θ_k, which is $\pi\sin(\tilde{\theta}_k)$ for half wavelength spacing and target angle $\tilde{\theta}_k$. Similarly, v_k is the speed induced phase shift of the k th target between two pulses, which is $2\pi\tilde{v}_kT_p/\lambda$ for target with a velocity of $\tilde{v}_k$.

Similarly, if the transmission sequence is changed to reverse, from antenna numbered $N-1$ to 0, the sampled signals are

$$\mathbf{s}_r[n]=\left[s_r[n,0],s_r[n,1],...,s_r[n,M-1]\right]^T, n=0,1,...,N-1, \quad (4)$$

where

$$\begin{aligned} s_r[n,m] &= \sum_{k=1}^{K}\sigma_{k,r}\exp\left(j\left(\left(N-1-n\right)M+m\right)\theta_k+jnv_k\right) \\ &= \sum_{k=1}^{K}\tilde{\sigma}_{k,r}\exp\begin{pmatrix} j\left(n-\left(N-1\right)/2\right)\left(-M\theta_k+v_k\right) \\ +j\left(m-\left(M-1\right)/2\right)\theta_k \end{pmatrix}, \end{aligned} \quad (5)$$

where the $\sigma_{k,r}$ and $\tilde{\sigma}_{k,r}$ are the signal complex amplitude of the k th target in reverse transmission. The signal amplitude remains unchanged, so $\left|\sigma_{k,f}\right|=\left|\sigma_{k,r}\right|$ but their phases are different and unknown due to different starting time of observation, which means that the forward and reverse signals are not coherent.

2.2 Proposed Joint Angle and Velocity Estimation

The column vectors $\mathbf{s}_f[n]$ can be arranged into a matrix

$$\begin{aligned} \mathbf{S}_f &= \left[\mathbf{s}_f[0],\mathbf{s}_f[1],...,\mathbf{s}_f[N-1]\right]^T \\ &= \begin{bmatrix} s_f[0,0] & s_f[0,1] & \cdots & s_f[0,M-1] \\ s_f[1,0] & s_f[1,1] & \cdots & s_f[1,M-1] \\ \vdots & \vdots & \ddots & \vdots \\ s_f[N-1,0] & s_f[N-1,1] & \cdots & s_f[N-1,M-1] \end{bmatrix}, \end{aligned} \quad (6)$$

where each target brings phase changes with radians of θ_k and $M\theta_k+v_k$ on the row and column respectively as shown in (2). So, the velocity and angle induced phases can be directly estimated by 2-dimensional Fourier transform of $\mathbf{S}_f$, where the phases estimated by finding peaks of the spectrum are represented by $\hat{\theta}_{k,f}$ and $\left(\widehat{M\theta_k+v_k}\right)_f$. $\hat{\theta}_{k,f}$ is used to unwrap $\left(\widehat{M\theta_k+v_k}\right)_f$ when the latter exceeds the range of $-\pi$ to π. An example of the 2-dimensinal spectrum of $\mathbf{S}_f$ is presented in Figure 1, together with the estimated phases. When the noise of each sample in $\mathbf{S}_f$ is independent, the estimations of the two parameters are also independent. In white Gaussian noise, when there is only one target or multiple targets with significant phase differences, the mutual influence between targets can be ignored and the estimation variance of $\hat{\theta}_{k,f}$ and $\left(\widehat{M\theta_k+v_k}\right)_f$ from 2-dimensional FFT is

$$\mathrm{var}\left(\hat{\theta}_{k,f}\right)=\frac{6\sigma_w^2}{N\left(M^3-M\right)\left|\sigma_{k,f}\right|^2}, \quad (7)$$

$$\mathrm{var}\left(\left(\widehat{M\theta_k+v_k}\right)_f\right)=\frac{6\sigma_w^2}{M\left(N^3-N\right)\left|\sigma_{k,f}\right|^2}, \quad (8)$$

where σ_w^2 is the noise power [21].

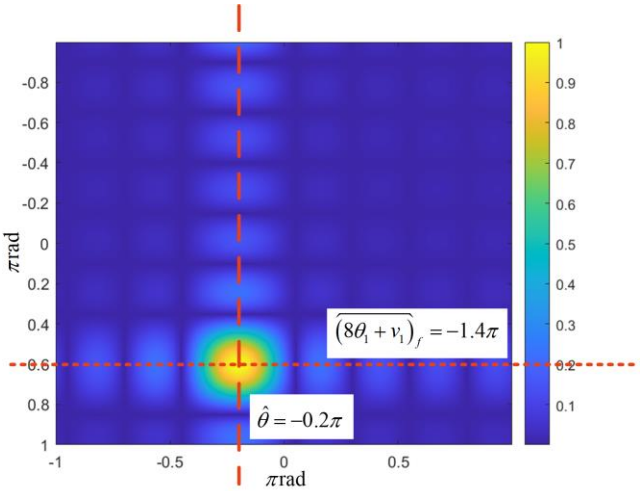


Fig. 1 2-dimensinal FFT of forward transmission matrix and estimated phase.

Similarly, by arranging the reverse transmitted signals $\mathbf{s}_r[n]$ into matrix $\mathbf{S}_r$ and performing two-dimensional Fourier

transform, estimations of θ_k and $-M\theta_k + v_k$ can be obtained and represented by $\hat{\theta}_{k,r}$ and $\overline{\left(-M\theta_k + v_k\right)}_r$ with similar estimation variance given in (7) and (8). The estimated results with similar angles $\hat{\theta}_{k,f}$ and $\hat{\theta}_{k,r}$ in forward and reverse transmission are paired and assigned the same subscript k. If the angles of multiple targets are indistinguishable, the results with the closest velocity derived from $\overline{\left(M\theta_k + v_k\right)}_f$ and $\overline{\left(-M\theta_k + v_k\right)}_r$ are paired.

With the four parameters of each target estimated using the forward and reverse transmission data, the target angle can be estimated by

$$\hat{\theta}_k = \frac{\left(\begin{array}{l} N\left(M^3 - M\right)\left(\hat{\theta}_{k,f} + \hat{\theta}_{k,r}\right) \\ +M^2\left(N^3 - N\right)\left(\overline{\left(M\theta_k + v_k\right)}_f - \overline{\left(-M\theta_k + v_k\right)}_r\right) \end{array}\right)}{2\left(M^3N^3 - MN\right)}, \quad (9)$$

where the four parameters are weighted averaged to obtained the final estimate of target angle $\hat{\theta}_k$. The weight of these parameters depends on their estimation variance under Gaussian white noise in (7) and (8).

Similarly, the estimate of velocity can be obtained as

$$\hat{v}_k = \frac{1}{2}\left(\overline{\left(M\theta_k + v_k\right)}_f + \overline{\left(-M\hat{\theta}_k + \hat{v}_k\right)}_r\right). \quad (10)$$

2.3 Algorithm Performance Analysis

Firstly, for single target or multiple targets with significant phase differences, the variance of $\hat{\theta}_k$ obtained from (9) is

$$\mathrm{var}\left(\hat{\theta}_k\right) = \frac{3\sigma_w^2}{\left(M^3N^3 - MN\right)\left|\sigma_{k,f}\right|^2}, \quad (11)$$

which is exactly the Cramer-Rao bound of target angle estimation based on signal model in (1) to (5), which is proved in the Appendix. In addition, (11) is the same as the estimation variance of MISO radar with MN receiving antennas and one transmitting antenna and two uncoherent transmissions, which has the same virtual aperture and free from coupling. So, the asymptotic accuracy of target angle estimation with proposed forward and reverse transmission and estimation method is not reduced with TDM system.

Similarly, the variance of velocity estimation by (10) is

$$\mathrm{var}\left(\hat{v}_k\right) = \frac{3\sigma_w^2}{M\left(N^3 - N\right)\left|\sigma_{k,f}\right|^2}, \quad (12)$$

which is also the Cramer-Rao bound of velocity estimation. So, for single or targets with significant parameter differences, the asymptotic performance of the proposed method under high signal-to-noise ratio (SNR) is optimal and can be given in closed form by (11) and (12).

Secondly, the resolution of the proposed method depends on whether the targets can be distinguished in forward or reverse transmission. The resolution is measured by the main lobe width of FFT. In forward transmission, two target 1 and 2 are distinguishable if

$$\left|\theta_1 - \theta_2\right| > 2\pi / M, \quad (13)$$

or

$$\left|M\theta_1 + v_1 - M\theta_2 - v_2\right| > 2\pi / N. \quad (14)$$

Similar, the two targets are distinguishable in reverse transmission if (13) is satisfied or

$$\left|v_1 - M\theta_1 - v_2 + M\theta_2\right| > 2\pi / N. \quad (15)$$

For closely spaced targets, the resolution mainly depends on (14) and (15), from which it can be derived that if

$$\left|\theta_1 - \theta_2\right| > 2\pi / \left(NM\right), \quad (16)$$

then the two targets are distinguishable regardless of their speed. It is also same as the resolution of MISO radar with similar aperture.

3 Simulations

3.1 Estimation Accuracy of Single Target

The estimation accuracy of the algorithm under white Gaussian noise is evaluated when there is only one target. Table 1 shows the simulation parameters, where the parameters of the first target with subscript 1 is used in this section. 10,000 Monte Carlo simulations are performed to get the root mean square errors (RMSE).

Angle estimations with MISO radar and random transmission TDM are used for comparison. The former uses one transmit antenna and $MN = 64$ receive antennas and transmits two pulses. The latter also uses the same 8 transmit antennas and 8 receive antennas as the proposed method, but the transmit antennas are activated in random order. Two pulses are transmitted by each transmit antenna, same as the proposed method. In both cases, matched filtering is used to estimate the angle, which should be optimal with single target [15].

Table 1 Parameters for simulation

Parameters	Value	Parameters	Value
M	8	N	8
θ_1	0	v_1	0.1π
$\sigma_{1,f}$	$\exp\left(0.1j\pi\right)$	$\sigma_{1,r}$	$\exp\left(0.3j\pi\right)$
θ_2	0.8π	v_2	-0.875π
$\sigma_{2,f}$	$\exp\left(-0.2j\pi\right)$	$\sigma_{2,r}$	$\exp\left(0.8j\pi\right)$
θ_3	-0.05π	v_3	0.1π
$\sigma_{3,f}$	$\exp\left(0.1j\pi\right)$	$\sigma_{3,r}$	$\exp\left(0.3j\pi\right)$

The simulated root mean squares of $\hat{\theta}_1$ are presented in Figure 2, together with the derived CRB in (11), which is same for all three method. The asymptotic performance of the three methods is also the same under high signal-to-noise ratio as expected, and the accuracy of proposed method does not decline with TDM and velocity coupling.

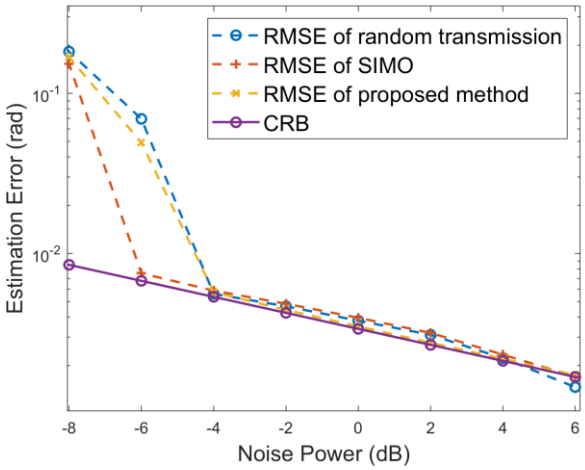


Fig. 2 Simulated root mean square error (RMSE) of single target angle estimation and the Cramer-Rao bound.

3.2 Estimation Performance of Multiple Target

Firstly, the second target in Table 1 is add to the simulation to evaluate the estimation performance with more than one target. The estimated RMSEs of angles and velocities of two targets are presented in Figure 3, together with the CRB given by (11) and (12). Though the derived CRB ignored the interaction between multiple targets, it still fits the simulated RMSEs well when the velocities and angles of the two targets differ greatly.

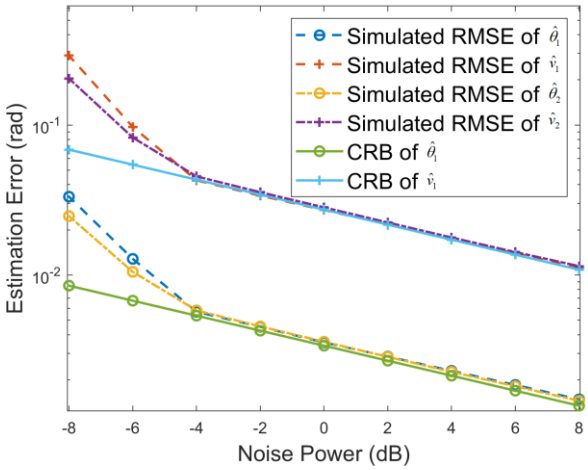
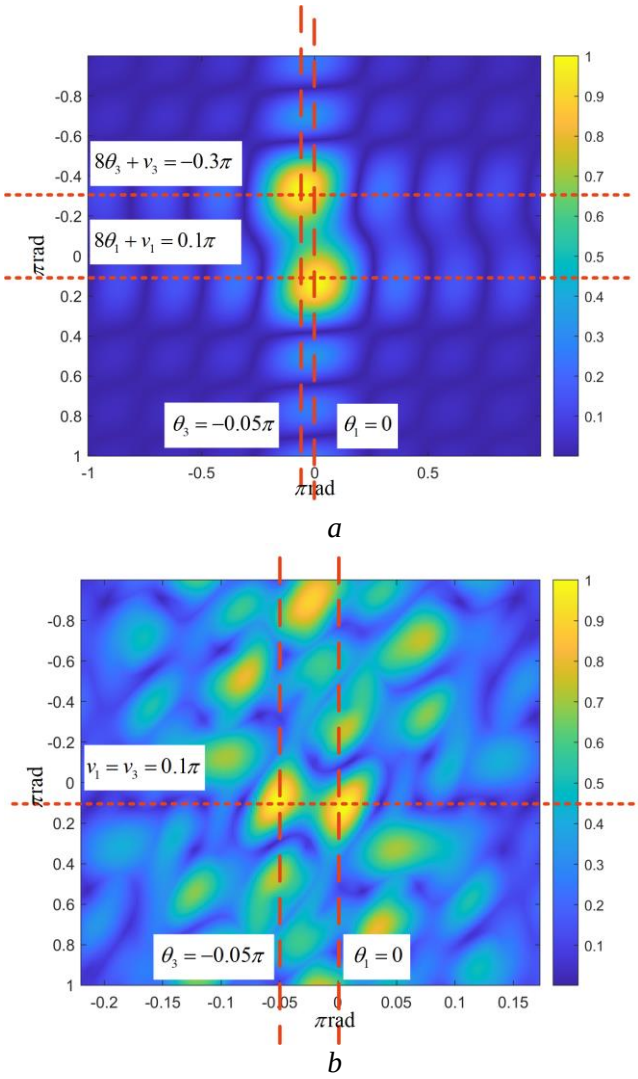


Fig. 3 Simulated root mean square error (RMSE) of two targets and the Cramer-Rao bound.

Secondly, the first and third targets in Table 1 with consistent speed and similar angle are used to evaluate the algorithm performance with two closely spaced targets. The 2-dimensinal spectrum of $\mathbf{S}_f$ is presented in Figure 4, together with the 2-D spectrum of random transmission with missing data filled by 0 and 2-D FFT [15]. Although the targets with an angle difference of 0.05 radians can be correctly distinguished in both cases, there are many side lobes in the latter which may handers the detection of the targets.



a

b

Fig. 4 (a) The 2-D spectrum from forward transmission of two targets with close angles, (b) The 2-D spectrum of random transmission.

4 Discussion

A new transmission pattern and target parameter estimation method for TDM radar is proposed and analysed in this paper. By activating the transmitting antenna forward and reverse, the phase changes caused by the target angle and velocity are superimposed or subtracted respectively and can be decoupled by solving the equations. The proposed method does not eliminate the coupling between velocity and angle in TDM radar, but utilize the coupled phase in forward and reverse transmission to simply cancel the mutual influence and obtain independent velocity and angle parameters.

Specifically, the proposed method has asymptotically optimal accuracy and low complexity. Its accuracy of single target estimation can be described by derived Cramer-Rao bound in close form, which is same as MISO array with same aperture. Due to the special arrangement of forward and reverse data matrixes, the algorithm only requires fast Fourier transform like traditional angle and velocity estimation methods. The minimum complexity of the algorithm is $O\left(MN\log_2\left(MN\right)\right)$

without FFT zero padding, which is smaller than the simplest matched filtering method of random transmission with a complexity of $O\left(MN^2 \log_2\left(MN^2\right)\right)$ [15]. Besides, there are no missing elements in the data matrix and the sampling is uniform, so there is no sidelobes in the 2-dimensional spectrum, which facilitates direct target detection and distinguish by peak finding. Further use of methods such as maximum likelihood or compressed sensing may help distinguish multiple targets, but at the cost of increasing complexity [18].

The main limitation of the proposed algorithm lies in the phase unwrapping and pairing of results from two estimations, which degenerates the algorithm performance with low signal-to-noise ratios. As shown in Figure 2, although the proposed method is asymptotically optimal, its performance deteriorated more rapidly at low SNR compared with the SIMO radar with matched filtering. The essence of unwrapping of $\overline{\left(M\theta_k + v_k\right)}_f$ and $\overline{\left(-M\theta_k + v_k\right)}_r$ by estimated $\hat{\theta}_{k,f}$ and $\hat{\theta}_{k,r}$ is to obtain the unwrapped estimate of v_k. If the target speed is far less than the maximum measurable range, the absolute value of v_k should be less than π and it can be easily unwrapped. On the contrary, unwrapping may be prone to error at low SNR. Similar problems also appear in traditional velocity estimation, but here it affects not only velocity but also the angle estimation.

The pairing is largely unavoidable as the two transmissions are not coherent. Future research will resort to target tracking and diversified waveforms to provide additional information of target angle and velocity and help results pairing. Besides, future research will also focus the speed estimation with coherent transmission and multiple forward and reverse pulse trains, which will provide better velocity resolution and help distinguish multiple targets in Doppler domain.

5 Conclusion

The proposed forward and reverse transmission pattern and parameter estimation method opens a new way to recover the target angle and velocity in TDM MIMO radar. Coupled phases are estimated from forward and reverse transmission data matrixes respectively and used to recover the target angle and velocity. Mathematical analysis and simulations show that the estimation accuracy and resolution of TDM radar with forward and reverse transmission are equivalent to MISO radar with the same aperture, completely eliminating the adverse influence of angle and velocity coupling.

6 Appendix

The Cramer-Rao bound of parameters estimation from (1) and (4) is presented below. Reformulate the signal into a column vector

$$\mathbf{s} = \left[\mathbf{s}_f^T[0],...,\mathbf{s}_f^T[N-1],\mathbf{s}_r^T[0],...,\mathbf{s}_r^T[N-1]\right]^T. \quad (17)$$

Unknown parameters include target angles

$$\boldsymbol{\theta} = \left[\theta_1,\theta_2,...,\theta_K\right]^T, \quad (18)$$

target velocities

$$\mathbf{v} = \left[v_1,v_2,...,v_K\right]^T, \quad (19)$$

And complex amplitudes

$$\boldsymbol{\sigma} = \left[\tilde{\sigma}_{1,f},...,\tilde{\sigma}_{K,f},\tilde{\sigma}_{1,r},...,\tilde{\sigma}_{K,r}\right]^T. \quad (20)$$

The Fisher matrix is

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_\sigma & \mathbf{J}_{\sigma\theta} & \mathbf{J}_{\sigma v} \\ \mathbf{J}_{\theta\sigma} & \mathbf{J}_\theta & \mathbf{J}_{\theta v} \\ \mathbf{J}_{v\sigma} & \mathbf{J}_{v\theta} & \mathbf{J}_v \end{bmatrix} \quad (21)$$

where the subscript indicates the parameters for partial derivation. Find the elements of each block matrix respectively

$$\begin{aligned}\left[\mathbf{J}_\theta\right]_{p,q} &= \frac{2}{\sigma_n^2}\mathrm{Re}\left\{\left(\frac{\partial \mathbf{s}}{\partial \theta_p}\right)^H\left(\frac{\partial \mathbf{s}}{\partial \theta_q}\right)\right\} \\ &= \frac{2}{\sigma_w^2}\mathrm{Re}\left\{\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\begin{pmatrix}\tilde{\sigma}_{p,f}^H\tilde{\sigma}_{q,f}d_f^2[m,n]\varphi_{p,q,f}[m,n] \\ +\tilde{\sigma}_{p,r}^H\tilde{\sigma}_{q,r}d_r^2[m,n]\varphi_{p,q,r}[m,n]\end{pmatrix}\right\},\end{aligned} \quad (22)$$

$$\begin{aligned}\left[\mathbf{J}_v\right]_{p,q} &= \frac{2}{\sigma_n^2}\mathrm{Re}\left\{\left(\frac{\partial \mathbf{s}}{\partial v_p}\right)^H\left(\frac{\partial \mathbf{s}}{\partial v_q}\right)\right\} \\ &= \frac{2}{\sigma_w^2}\mathrm{Re}\left\{\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\begin{pmatrix}\tilde{\sigma}_{p,f}^H\tilde{\sigma}_{q,f}\varphi_{p,q,f}[m,n]t^2[m,n] \\ +\tilde{\sigma}_{p,r}^H\tilde{\sigma}_{q,r}\varphi_{p,q,r}[m,n]t^2[m,n]\end{pmatrix}\right\},\end{aligned} \quad (23)$$

where $p,q \in \left[1,2,...,K\right]$ and

$$d_f[m,n] = nM + m - NM/2 + 1/2, \quad (24)$$

$$d_r[m,n] = -nM + m + NM/2 - M + 1/2, \quad (25)$$

$$t[m,n] = n - \left(N-1\right)/2, \quad (26)$$

and

$$\begin{cases}\varphi_{p,q,f}[m,n] = \exp\left(jd_f[m,n]\left(\theta_q - \theta_p\right) + jt[m,n]\left(v_q - v_p\right)\right) \\ \varphi_{p,q,r}[m,n] = \exp\left(jd_r[m,n]\left(\theta_q - \theta_p\right) + jt[m,n]\left(v_q - v_p\right)\right)\end{cases}. \quad (27)$$

If there is only one target, or multiple targets with significant angle and velocity differences, which makes the summations include $\varphi_{p,q,f}[m,n]$ and $\varphi_{p,q,r}[m,n]$ with $p \neq q$ approximately 0 [22], then all elements in $\mathbf{J}_{\sigma\theta}$, $\mathbf{J}_{\sigma v}$, $\mathbf{J}_{\theta\sigma}$ and $\mathbf{J}_{v\sigma}$ are 0 and $\mathbf{J}_\theta$ and $\mathbf{J}_v$ are diagonal. Additionally, if $\left|\sigma_{k,f}\right| = \left|\sigma_{k,r}\right|$ as assumed in Section 2, $\mathbf{J}_{\theta v}$ and $\mathbf{J}_{v\theta}$ are 0 and the inverse of Fisher matrix is

$$\mathbf{J} = \mathrm{diag}\left(\mathbf{J}_\sigma^{-1},\mathbf{J}_\theta^{-1},\mathbf{J}_v^{-1}\right), \quad (28)$$

Where the diag() forms diagonal matrix by the elements in brackets. The minimum estimation variance of $\hat{\theta}_k$ and $\hat{v}_k$ are

$$\begin{aligned}\left[\mathbf{J}_\theta\right]_{k,k}^{-1} &= \frac{\sigma_w^2}{2\left|\tilde{\sigma}_{k,f}\right|^2}\left(\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\left(d_f^2[m,n] + d_r^2[m,n]\right)\right)^{-1} \\ &= \frac{3\sigma_w^2}{\left(M^3N^3 - MN\right)\left|\sigma_{k,f}\right|^2}\end{aligned} \quad (29)$$

$$\left[\mathbf{J}_v\right]_{k,k}^{-1} = \frac{\sigma_w^2}{4\left|\tilde{\sigma}_{k,f}\right|^2}\left(\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\left(t^2[m,n]\right)\right)^{-1} = \frac{3\sigma_w^2}{M\left(N^3-N\right)\left|\sigma_{k,f}\right|^2} \tag{30}$$

Which are (11) and (12), respectively.

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